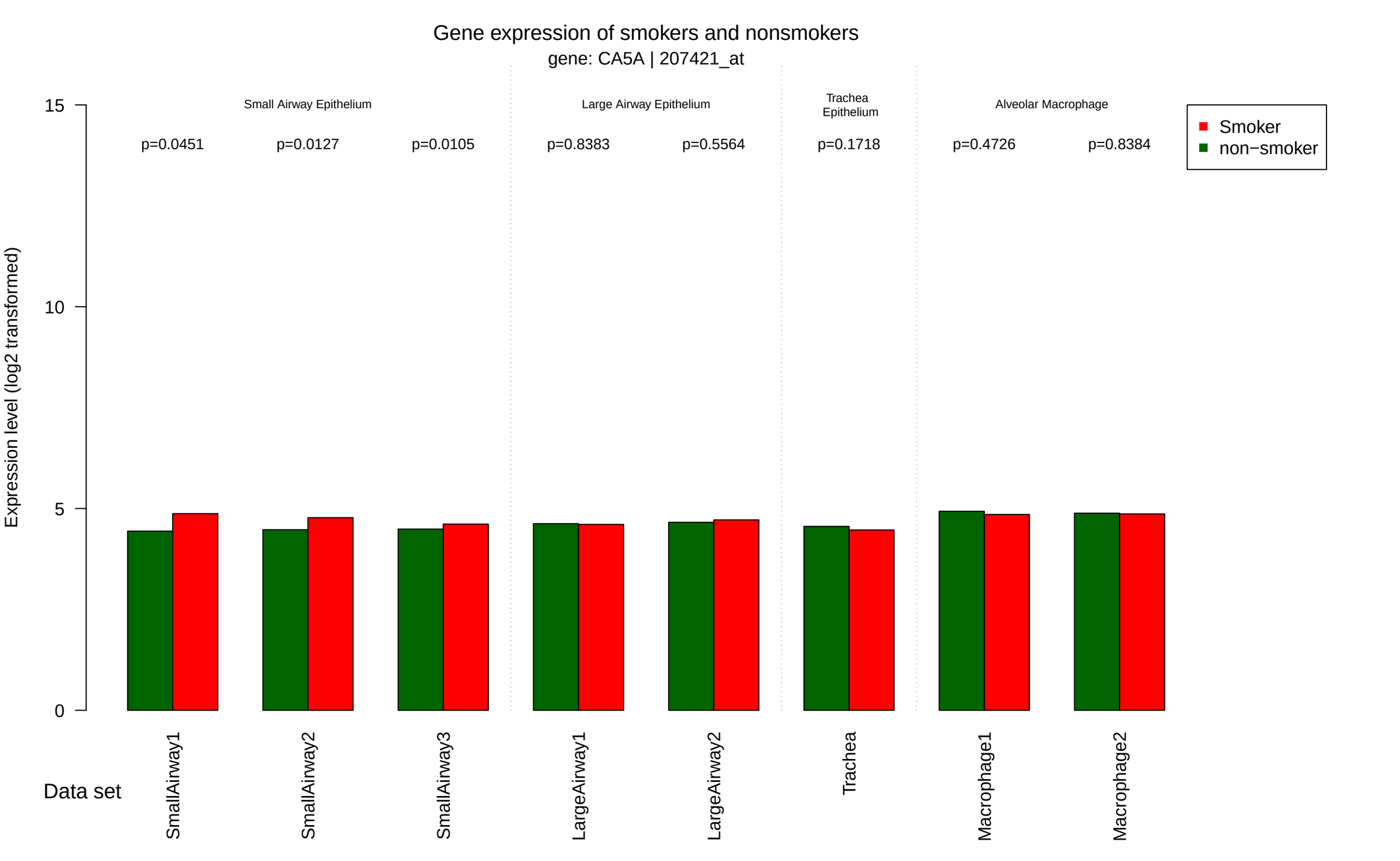
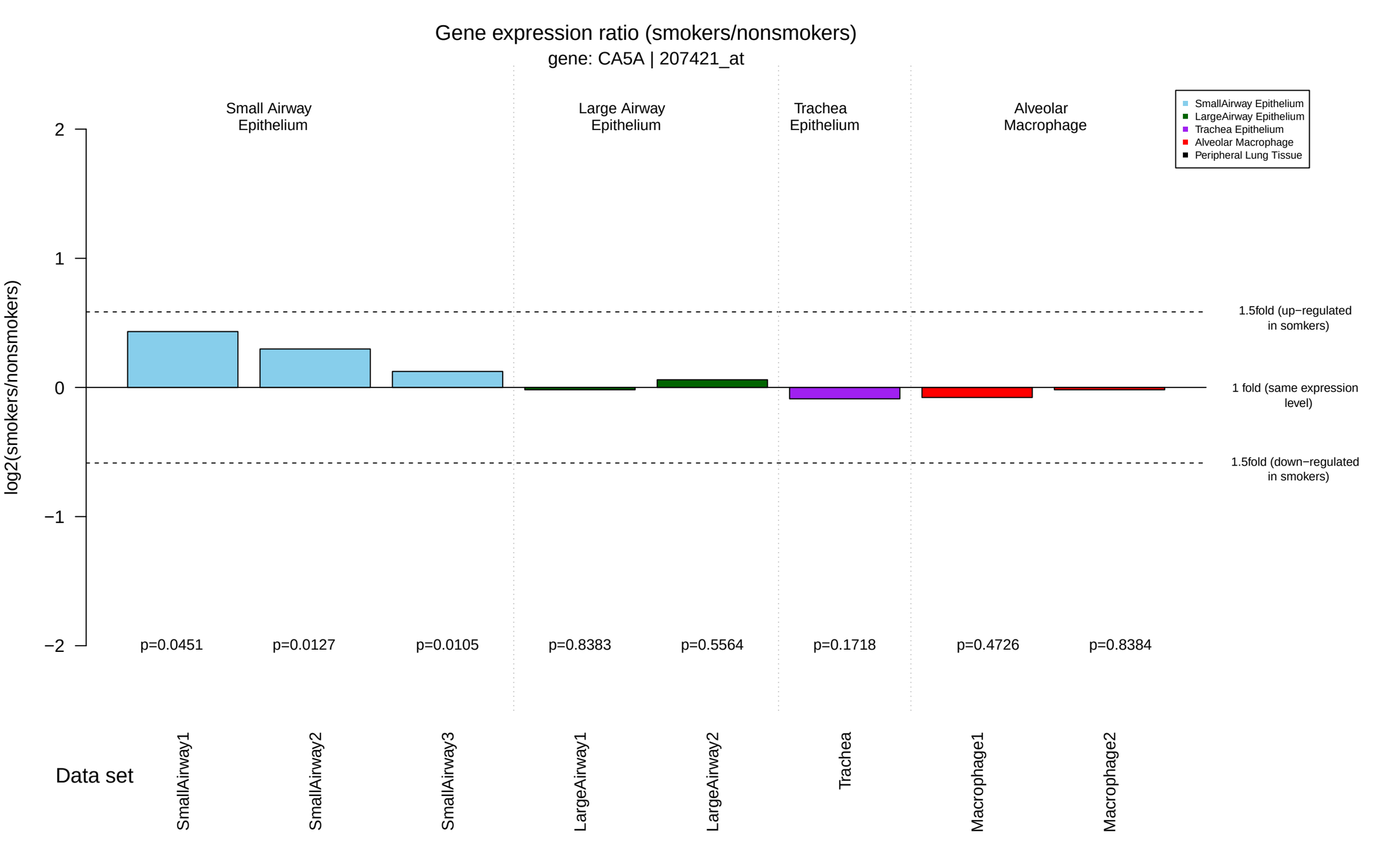
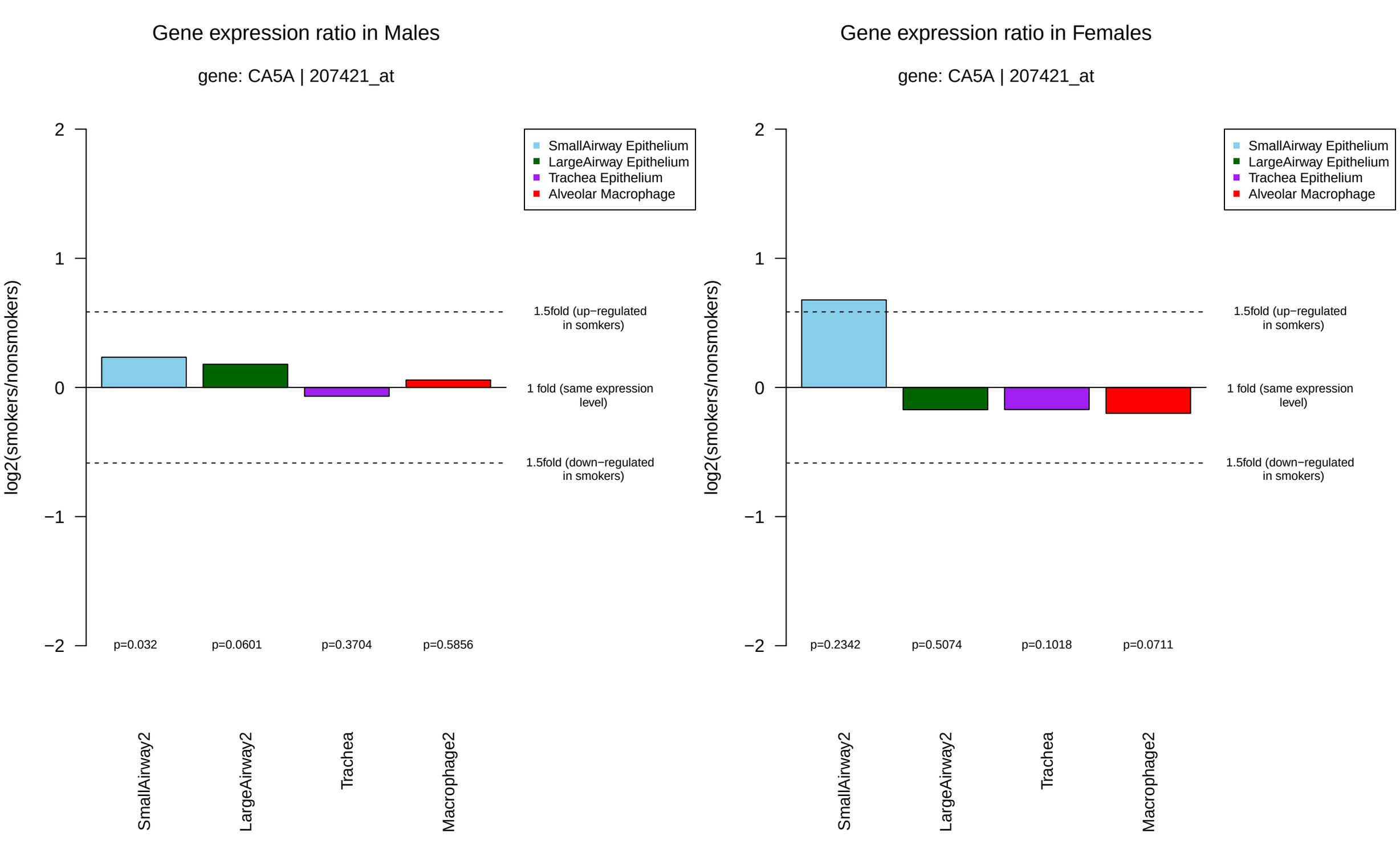
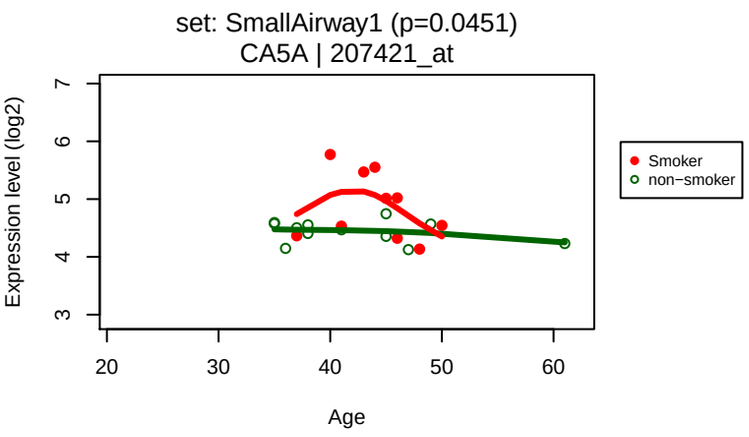
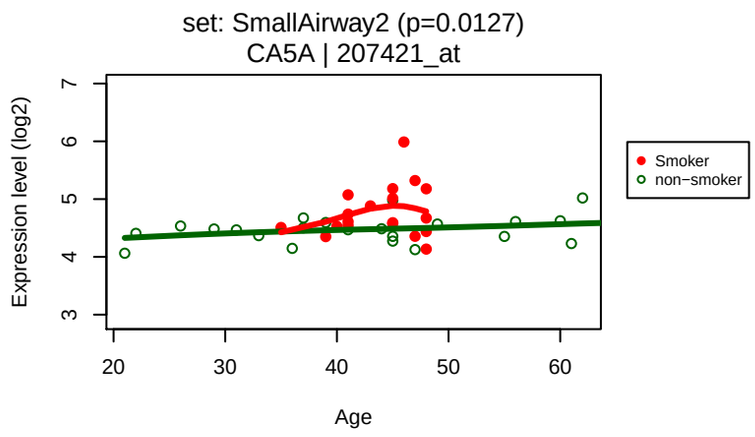
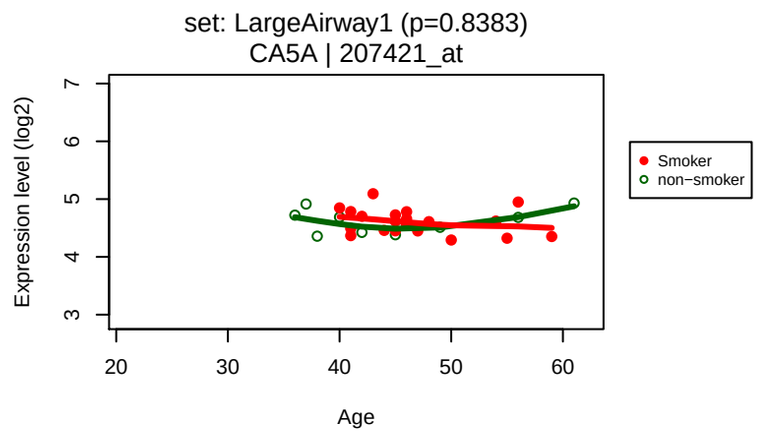

Gene expression of smokers and nonsmokers
gene: CA5A | 207421_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0451
p=0.0127
p=0.0105
p=0.8383
p=0.5564
p=0.1718
p=0.4726
p=0.8384
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

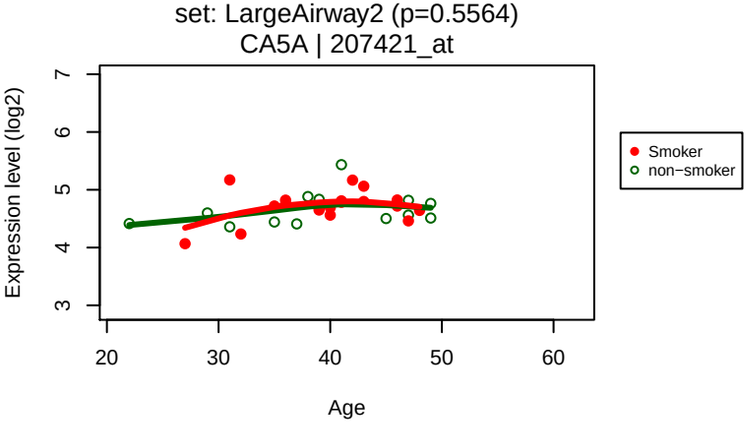

Gene expression ratio (smokers/nonsmokers)
gene: CA5A | 207421_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0451
p=0.0127
p=0.0105
p=0.8383
p=0.5564
p=0.1718
p=0.4726
p=0.8384
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

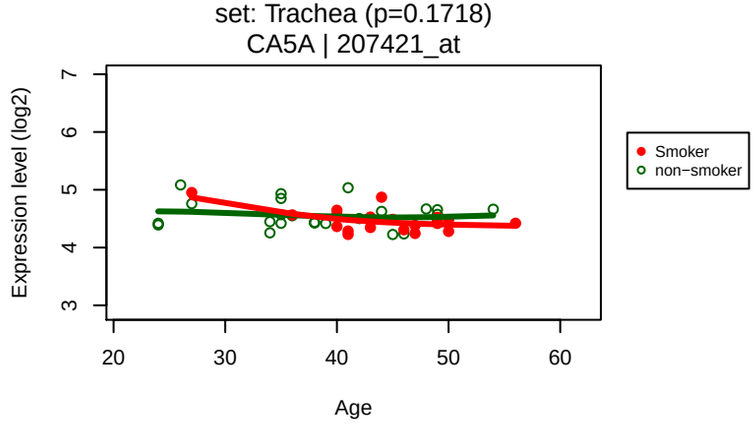

Gene expression ratio in Males
gene: CA5A | 207421_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.032
p=0.0601
p=0.3704
p=0.5856
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CA5A | 207421_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2342
p=0.5074
p=0.1018
p=0.0711
SmallAirway2
LargeAirway2
Trachea
Macrophage2

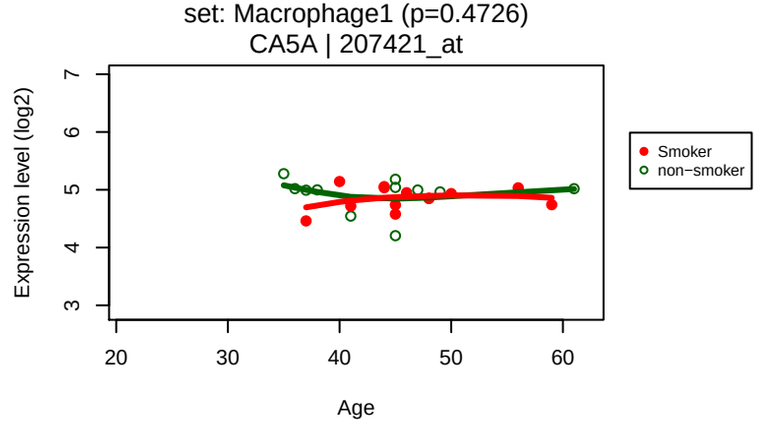

set: SmallAirway1 (p=0.0451)
CA5A | 207421_at
Expression level (log2)
Age
Smoker
non−smoker

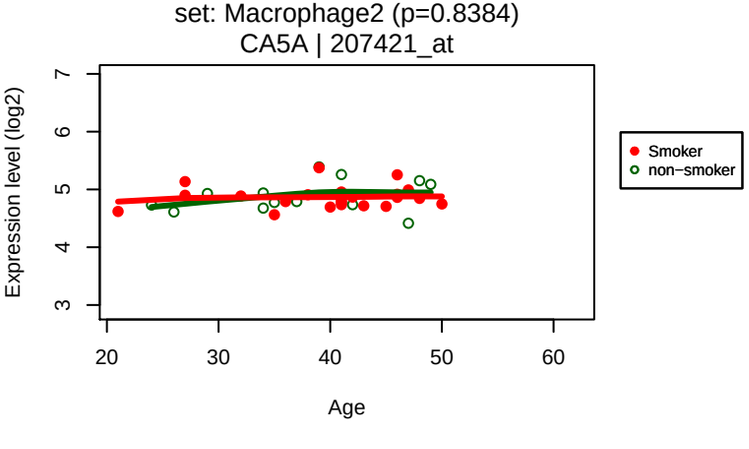

set: SmallAirway2 (p=0.0127)
CA5A | 207421_at
Expression level (log2)
Age
Smoker
non−smoker

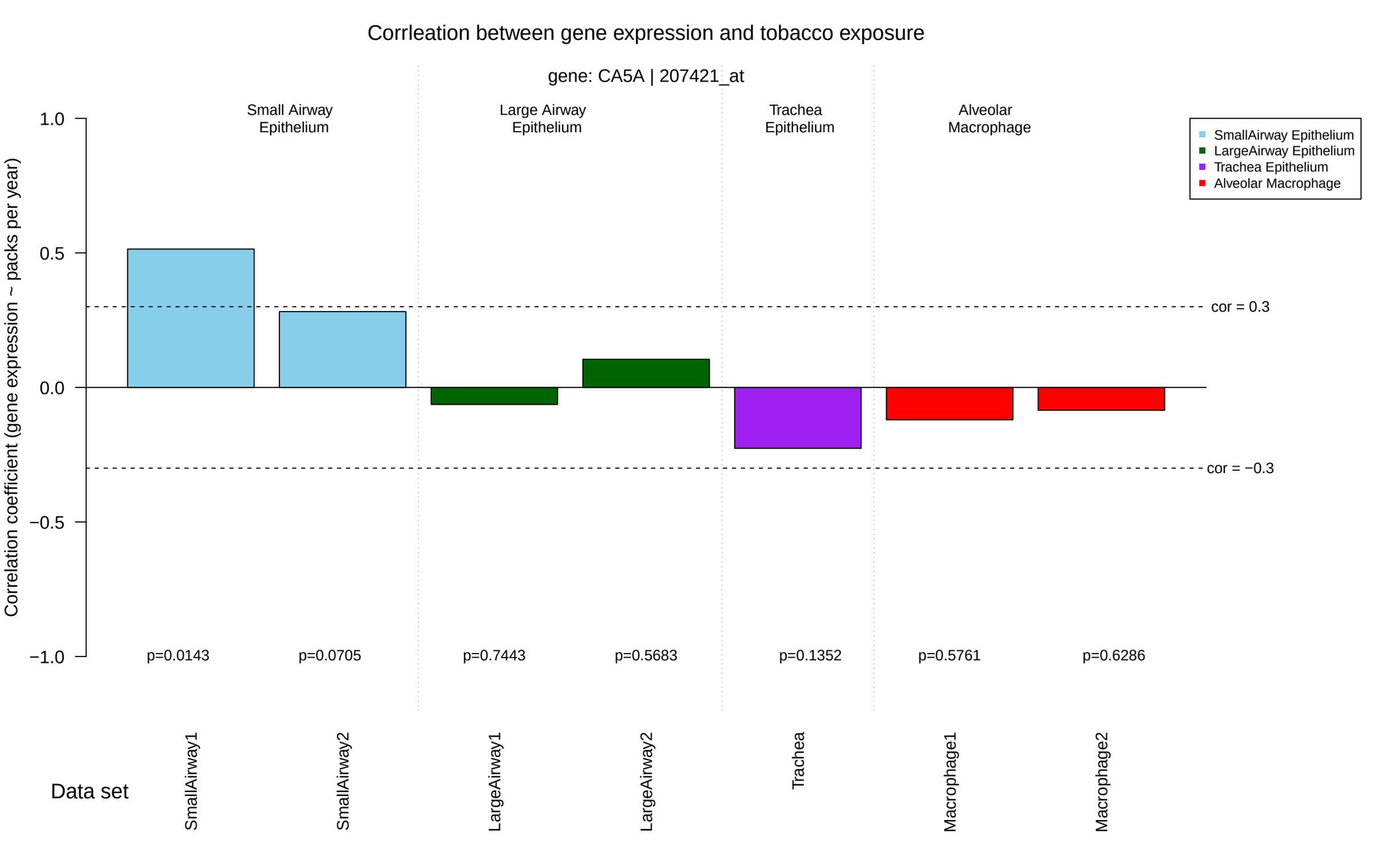

set: LargeAirway1 (p=0.8383)
CA5A | 207421_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5564)
CA5A | 207421_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1718)
CA5A | 207421_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4726)
CA5A | 207421_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8384)
CA5A | 207421_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CA5A | 207421_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0143
p=0.0705
p=0.7443
p=0.5683
p=0.1352
p=0.5761
p=0.6286
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2